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Credit Portfolio Model With Climate Overlay

A factor-based framework with deterministic climate/scenario drivers and systematic credit factors with market-calibrated correlation structure.

Related Documentation: - **Implementation Plan**: `/docs/implementation_plan.md` - **Data Sources**: `/docs/data_sources.md` (all external data - scenarios, correlations, calibration) - **Calibration Parameters**: `/data/calibration/README.md` - **Quick Reference**: `/docs/correlation_calibration_quick_reference.md`

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1. Purpose and scope

What we are trying to achieve

We want a credit portfolio model that produces a distribution of defaults and losses under a chosen deterministic climate scenario, while preserving familiar factor-model behaviour (correlated outcomes, portfolio tails) using a structured, calibratable correlation layer.

The model should:

1. Behave like a standard credit factor model (familiar to risk practitioners and governance).
2. Translate climate scenario variables into systematic credit stress in a transparent, auditable way.
3. Support scenario analysis and stress testing rather than claim to “predict” climate-driven defaults from first principles.
4. Be implementable even when we lack clean historical observables to statistically estimate climate-credit parameters.

Key simplification we adopt

We keep one main factor object that drives correlated credit outcomes: - **F**: a sector–region factor matrix, representing systematic “credit conditions” for each sector and region cell.

Climate enters through: - **X**: a deterministic vector of climate/scenario variables (e.g., carbon price path, energy prices, hazard intensities), which drives F via a mapping $F = m(X) + u$.

So the model is: - Credit outcomes depend on F - F = m(X) + u where m(X) is deterministic scenario mean, u is stochastic residual - Idiosyncratic noise provides name-level randomness.

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2. Conceptual overview

Three layers of the system

1. **Scenario layer (deterministic)**: A climate scenario gives a set of driver values X (e.g., carbon price change, energy price shock, physical hazard indices). These are **not random** in the model.

2. **Systematic credit layer (stochastic)**: The scenario shifts the mean systematic credit conditions for each sector × region cell. Around that mean, there is a stochastic residual credit environment that creates correlation and tail risk.

3. **Obligor layer (stochastic)**: Each obligor loads on the sector × region systematic credit conditions via weights β_i and has an idiosyncratic shock. A single "asset-correlation" knob ϱ controls how strongly obligors co-move with systematic conditions overall.

Why a sector-region cell structure?

The cell-level factor matrix $F_{s,r}$ provides: - **Natural interpretation**: $F_{coal,europe}$ directly represents "coal sector in Europe" - **Flexible exposures**: Obligors can load on multiple cells (diversified portfolios) - **Climate overlay**: Deterministic scenario X drives systematic mean $m_{s,r}(X)$ - **Residual correlation**: Stochastic u captures non-climate systematic uncertainty via market-calibrated covariance Σ_u

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3. Underlying theory: credit portfolio factor models (in plain terms)

The classic structure

Most portfolio credit models start by constructing a latent creditworthiness variable for each obligor i:

```
Z_i = systematic part + idiosyncratic part
```

The systematic part is shared across obligors (causing correlated outcomes). The idiosyncratic part is independent.

A common, governance-friendly form is:

```
Z_i = √ρ · (systematic score) + √(1−ρ) · ε_i
```

- **$\varrho \in (0,1)$**: fraction of variance that is systematic ("asset correlation")

- **ε_i**: idiosyncratic noise (standard normal)

A default event is triggered by a threshold:

```
Default if Z_i < τ_i
```

The threshold τ_i is set to reproduce the obligor's unconditional probability of default (PD), typically via:

```
τ_i = Φ^(-1)(PD_i)
```

where Φ is the standard normal CDF.

Why this works

- The model reproduces marginal PDs by construction.
- It generates default correlation through shared systematic component.
- It can be simulated efficiently (Monte Carlo) and decomposed by risk drivers.
- The single parameter ϱ controls tail thickness and diversification benefits.

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4. Our model specification

4.1 Index sets and dimensions

Sectors: s = 1,…,S (e.g., S=15) **Regions**: r = 1,…,R (e.g., R=7) **Obligors**: i = 1,…,N **Climate drivers**: k = 1,…,K (K=8 in our implementation)

Number of sector–region cells:

```
M := S × R    (e.g., M = 105 for 15 sectors × 7 regions)
```

Indexing convention: Sector-major (regions vary fastest within sector)

```
j = idx(s,r) := (s-1)R + r
```

So j=1 is (s=1, r=1), j=2 is (s=1, r=2), …, j=R is (s=1, r=R), j=R+1 is (s=2, r=1), etc.

4.2 Factor surface definition

Define the systematic factor matrix:

```
F ∈ ℝ^(S×R)
```

Where F_{s,r} represents the systematic credit condition for sector s in region r.

Vector form:

```
f := vec(F) ∈ ℝ^M
```

Interpretation: - $F_{coal,europe}$ represents systematic credit stress for coal companies in Europe - $F_{agriculture,asia}$ represents systematic credit stress for agriculture in Asia - These are **latent random variables** conditional on climate scenario

4.3 Factor decomposition: Scenario mean + residual

Each cell factor is decomposed as:

$$f = m(X) + u$$

Where: - $\mathbf{m(X)} \in \mathbb{R}^{\mathbf{M}}$: Deterministic scenario-driven mean (climate overlay) - $\mathbf{u} \in \mathbb{R}^{\mathbf{M}}$: Stochastic residual ~ $N(0, \Sigma_u)$ (non-climate systematic uncertainty)

Key insight: Climate fixes the mean; Σ_u controls correlated randomness around that mean.

4.4 Obligor exposure weights (must sum to 1)

Each obligor i has exposure weights:

$$\beta_i \in \mathbb{R}^M$$

With the constraint:

$$\beta_{i,j} \geq 0 \quad \text{for all } j, \qquad \Sigma_j \beta_{i,j} = 1$$

So β_i is a **convex combination** across sector–region cells.

Single-cell assignment (special case): - If obligor i belongs purely to cell (s(i), r(i)), then β_i has a 1 in position idx(s(i),r(i)) and 0 elsewhere.

Multi-cell assignment (diversified exposure): - If obligor i has 60% revenue from coal in Europe, 40% from oil in North America: - $\beta_{idx(coal,europe)} = 0.6$, $\beta_{idx(oil,NA)} = 0.4$, all others = 0

Data source: Financial reports, segment disclosures, asset locations

4.5 Obligor latent variable with asset correlation ϱ

We define the obligor's systematic score:

$$S_i := \beta_i^T \cdot f$$

Since different β_i imply different systematic variance, we **standardize**:

$$\mathrm{Var}(S_i) = \beta_i^T \Sigma_u \beta_i$$

$$\tilde{S}_i := S_i / \sqrt{\beta_i^T \Sigma_u \beta_i}$$

Now $\tilde{S}_i$ has unit variance.

Then we introduce a single global systematic share $\boldsymbol{\varrho} \in \mathbf{(0,1)}$:

```
Z_i = √ρ · S̃_i + √(1−ρ) · ε_i,    ε_i ~ N(0,1)
```

Interpretation of ϱ: - ϱ is the fraction of variance in Z_i that is systematic for every obligor. - Higher ϱ ⇒ stronger default clustering and fatter portfolio tails. - Typical range: $\varrho \in [0.15, 0.25]$ calibrated to match portfolio tail behavior.

4.6 Default and loss

Default threshold matches baseline PD:

```
Default_i = 1{Z_i < Φ^(−1)(PD_i)}
```

Loss:

```
L_i = Default_i · EAD_i · LGD_i

L = Σ_i L_i
```

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5. Climate overlay: deterministic scenario drivers X

5.1 Scenario driver vector X

X is a vector of scenario variables:

```
X ∈ ℝ^K    (K = 8 in our implementation)
```

It is **deterministic** for a chosen scenario and horizon. For NGFS "Net Zero 2050" year 2050, X is fixed.

Deterministic scenario assumption:

```
X = μ^(scenario)
```

No randomness is introduced at the climate-driver level.

Transition drivers (5): 1. CarbonPrice - Carbon tax/price (US$2010/t CO2) 2. CoalPrice - Coal price index (US$2010/GJ) 3. OilPrice - Oil price index (US$2010/GJ) 4. GasPrice - Natural gas price (US$2010/GJ) 5. GDP - GDP growth/deviation (billion US$2010/yr, PPP)

Physical drivers (3): 6. HeatIndex - Days with heat index >35°C 7. FloodRisk - Maximum 1-day precipitation (mm) 8. DroughtRisk - Consecutive dry days

Crucially: X does not directly appear in the obligor equation. It determines the **mean** of systematic factors: m(X).

5.2 Standardization of drivers ϕ(X)

We standardize drivers to unitless "sigma" terms:

$$\varphi_k(X) = \Delta X_k / \sigma_k$$

Where: - **ΔX_k**: Scenario change for driver k relative to baseline (e.g., Current Policies) - **σ_k**: Scale parameter ("typical move" for driver k)

In our implementation: - **σ_k = cross-scenario standard deviation** (computed from Net Zero, Delayed, Current Policies) - **ϕ_k ranges from -2σ to +2σ** across scenarios

Interpretation: $\phi_k = 1$ means "a 1-sigma move in driver k relative to baseline"

5.3 Climate-to-credit mapping m(X)

The deterministic climate mean is a linear function of standardized drivers:

Cellwise:

$$m_{s,r}(X) = \alpha_{s,r} + \Sigma_k w_{s,r,k} \cdot \varphi_k(X)$$

Vector form:

$$m(X) = \alpha + W \cdot \varphi(X)$$

Where: - **$\alpha := \mathrm{vec}([\alpha_{s,r}]) \in \mathbb{R}^M$**: Baseline intercept (typically 0 if F represents deviations) - **$W \in \mathbb{R}^{(M \times K)}$**: Sensitivity matrix ("how driver k affects cell j")

Interpretation: $W\,\phi(X)$ is the scenario-implied shift in credit conditions across sector×region cells.

5.4 Optional "hybrid" structured parameterization for W

Rather than calibrate M×K = 840 weights independently, you can use structured parameterization:

Step 1: Define relative exposure scores - **$S_{s,k} \in [0,1]$**: How exposed sector s is to driver k - **$R_{r,k} \in [0,1]$**: How exposed region r is to driver k

Step 2: Combine via product

$$w_{(s,r),k} = \lambda_k \cdot S_{s,k} \cdot R_{r,k}$$

Where: - **λ_k**: Overall scale for driver k

Interpretation: - **S and R** encode relative vulnerabilities (which sectors/regions are most exposed) - **λ_k** controls overall magnitude (how strongly driver k translates to credit stress)

This reduces calibration from 840 parameters to: - S matrix: 15 sectors × 8 drivers = 120 scores - R matrix: 7 regions × 8 drivers = 56 scores - λ vector: 8 scale parameters - **Total: 184 parameters** (vs 840), with clear economic interpretation

Note: This is **optional but governance-friendly**. The fundamental object is W; the hybrid structure is a convenient parameterization.

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6. Residual systematic uncertainty: Σ_u and correlation structure

6.1 The residual covariance matrix

The stochastic residual captures non-climate systematic uncertainty:

```
u ~ N(0, Σ_u)
```

Where **Σ_u ∈ ℝ^(M×M)** is the residual covariance matrix that encodes: - Inter-sector co-movement - Inter-region co-movement - Portfolio tail thickness (together with ϱ) - Diversification benefits

Key design principle: We **cannot estimate climate-credit correlations robustly** from historical data. Instead, we: - Keep climate deterministic (no correlation parameters needed) - Put stochastic correlations into Σ_u - **Calibrate Σ_u from observable market data** (equity sector/region indices)

6.2 Practical construction of Σ_u from market data

Instead of a full free M×M matrix (5,460 parameters), build it from sector and region correlation blocks:

Step 1: Estimate sector and region correlations from market data

- **Corr_S**: Sector correlation matrix (S×S) estimated from sector index returns (e.g., MSCI sector indices)
- **Corr_R**: Region correlation matrix (R×R) estimated from regional index returns (e.g., MSCI regional indices)

Step 2: Convert to covariances with chosen volatility scales

```
Σ_S = D_S · Corr_S · D_S
Σ_R = D_R · Corr_R · D_R
```

Where D_S and D_R are diagonal matrices with sector/region volatilities (often start with D_S = D_R = I for simplicity).

Step 3: Embed into cell space using membership matrices

Define **membership matrices**: - **A ∈ ℝ^(M×S)**: Maps each cell to its sector - A_{j,s} = 1 if cell j belongs to sector s, else 0 - **B ∈ ℝ^(M×R)**: Maps each cell to its region - B_{j,r} = 1 if cell j belongs to region r, else 0

Step 4: Mix sector and region components with ω

Define **sector-vs-region mix parameter ω ∈ [0,1]**: - ω controls whether residual co-movement is primarily sector-cycle or region-cycle driven - High ω (→ 1): Sector cycles dominate (e.g., energy sector co-moves globally) - Low ω (→ 0): Regional cycles dominate (e.g., geographic macro conditions matter more) - Typical starting point: ω = 0.5 (equal weight)

Construct the structured covariance component:

```
Σ_structured = ω · (A Σ_S A^T) + (1−ω) · (B Σ_R B^T)
```

Step 5: Mix structured vs cell-specific with η

Define **structured-vs-cell share parameter η ∈ [0,1]**: - η controls how much of residual variance is structured (via Σ_structured) vs cell-specific - High η (→ 1): Strong systematic clustering, less diversification benefit - Low η (→ 0): More cell-specific noise, greater diversification benefit - Typical starting point: η = 0.7 (moderately systemic)

Construct final residual covariance:

```
Σ_u = η · Σ_structured + (1−η) · I_M
```

Expanding:

```
Σ_u = η · [ω · (A Σ_S A^T) + (1−ω) · (B Σ_R B^T)] + (1−η) · I_M
```

Knobs: - **Corr_S, Corr_R**: Estimated from data (MSCI sector/regional indices, 5-year rolling correlations) - **D_S, D_R**: Volatility scales (typically set to I for unit volatility) - **ω**: Sector-vs-region mix (0 = all region, 1 = all sector, 0.5 = equal) - **η**: Structured-vs-cell share (0 = all cell-specific, 1 = all structured)

6.3 What these components do

Corr_S (sector correlation): - Captures how different sectors co-move (e.g., coal and oil&gas move together due to energy policy) - Typical values: 0.3-0.5 for related sectors, 0.1-0.2 for unrelated sectors - Estimated from MSCI Global Sector Indices or credit sector sub-indices

Corr_R (region correlation): - Captures how different regions co-move (e.g., Europe and Asia linked by trade) - Typical values: 0.2-0.4 for developed regions, 0.3-0.5 for emerging regions - Estimated from MSCI Regional Indices or regional GDP growth correlations

D_S, D_R (volatility scales): - Controls overall strength of sector/region components - Typically set to I (unit volatility) for simplicity - Alternative: Use empirical std devs if sector/region-specific volatilities are desired

ω (sector-vs-region mix): - Controls whether residual co-movement is driven by sector cycles or regional cycles - ω = 1: Pure sector co-movement (coal in Europe and coal in Asia move together) - ω = 0: Pure regional co-movement (coal in Europe and oil in Europe move together) - ω = 0.5: Balanced mix of both - **Calibration**: Start with 0.5, adjust based on portfolio composition and risk decomposition - **Sensitivity**: Test ω ∈ {0.3, 0.5, 0.7} to understand sector vs region dominance

η (structured-vs-cell share): - Controls how much residual variance is systematic vs cell-specific - η = 1: All residual variance is structured (maximum clustering, minimum diversification) - η = 0: All residual variance is cell-specific (maximum diversification, no clustering) - η = 0.7: 70% structured, 30% cell-specific (moderately systemic) - **Calibration**: Start with 0.7, adjust to match observed tail thickness and diversification - **Sensitivity**: Test η ∈ {0.5, 0.7, 0.9} to understand impact on VaR and ES - **Economic interpretation**: η is like an “intra-cell correlation” - higher η means cells are more correlated beyond sector/region patterns

6.4 Induced correlation structure

The construction $\Sigma_u = \eta \cdot [\omega \cdot (A \Sigma_S A^T) + (1-\omega) \cdot (B \Sigma_R B^T)] + (1-\eta) \cdot I$ induces:

Within-sector correlation (same sector, different regions):

```
Corr(u_{s,r}, u_{s,r'}) = (η · ω · Sector variance component) / (Total
variance)
```

- Controlled by η (strength of structure) and ω (sector vs region weight)
- Higher η and ω → stronger within-sector correlation

Within-region correlation (different sectors, same region):

```
Corr(u_{s,r}, u_{s',r}) = (η · (1−ω) · Region variance component) /
(Total variance)
```

- Controlled by η and $(1-\omega)$
- Higher η and lower ω → stronger within-region correlation

Cross-sector-region correlation (different sector AND region):

```
Corr(u_{s,r}, u_{s',r'}) ≈ 0  (by construction, no direct linkage)
```

Diagonal variance (variance of each cell):

```
Var(u_{s,r}) = η · [ω · (A Σ_S A^T)_{s,r} + (1−ω) · (B Σ_R B^T)_{s,r}]
+ (1−η)
```

- With $D_S = D_R = I$, the diagonal is approximately $\eta + (1-\eta) = 1$ (unit variance cells)

Key advantages: - **Observable data**: Corr_S and Corr_R estimated from equity/credit market indices - **Parsimonious**: Only 4 key knobs (ω, η, plus Corr_S and Corr_R structures) - **Interpretable**: ω controls sector vs region, η controls clustering strength - **Separable**: Can tune sector-region balance (ω) independently from overall clustering (η) - **Stable simulation**: Well-conditioned by construction, positive definite

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7. The correlation levers (intuitive summary)

There are five distinct "correlation levers" and they do different jobs:

Lever 1: Corr_S and Corr_R (correlation patterns) - "Which sectors co-move? Which regions co-move?" - Estimated from market data (MSCI sector/regional indices) - Provides the underlying correlation structure

Lever 2: ω (sector-vs-region mix) - "Is residual co-movement driven by sector cycles or regional cycles?" - $\omega = 1$: Sector dominates (e.g., global energy sector cycle) - $\omega = 0$: Region dominates (e.g., regional macro conditions) - Typical starting point: $\omega = 0.5$ (balanced)

Lever 3: η (structured-vs-cell share) - "How much residual variance is systematic vs cell-specific?" - Higher η ⇒ stronger clustering, less diversification, fatter tails - Lower η ⇒ more cell-specific noise, more diversification - Typical starting point: η = 0.7 (moderately systemic)

Lever 4: ϱ (asset correlation at obligor level) - "How strongly do obligors respond to systematic conditions vs idiosyncratic noise?" - Higher ϱ ⇒ stronger default clustering, fatter tails - Lower ϱ ⇒ more idiosyncratic behavior, thinner tails - Typical range: ϱ ∈ [0.15, 0.25]

Lever 5: β_i (who is exposed to which cells) - "Which obligors are linked to which sector×region conditions?" - From portfolio data (sector/region assignments, revenue splits)

Climate scenarios do not appear as correlations; they appear as mean shifts m(X).

That keeps the model auditable: - Climate scenario assumptions are fully visible in X, σ_k, and W - Correlation assumptions are fully visible in Corr_S, Corr_R, ω, η, and ϱ

---

8. The calibration "knobs": what you must choose/estimate

Think of knobs in four groups.

A) Scenario knobs (deterministic)

(A1) Scenario selection - **Knob**: Choice of scenario (NGFS pathway, internal pathway) - **Effect**: Sets X = μ^(scenario) and therefore the mean stress surface m(X)

(A2) Driver definitions (X_k) - **Knob**: Which drivers you include (carbon price, energy price, flood index, etc.) - **Effect**: Determines what kinds of climate effects you can express - **Design note**: Can start "global" and later add more drivers without changing engine

(A3) Feature scaling σ_k - **Knob**: "Typical move" for each driver - **Effect**: Defines how big ϕ_k(X) is; scales scenario impact for a given W - **Practical role**: Interpretability ("1σ carbon shock") and stability across scenarios

B) Climate→credit transmission knobs (mean mapping)

(B1) Sensitivity matrix W (or structured scores) - **Knob**: W_{(s,r),k} or the score libraries S_{s,k}, R_{r,k} - **Effect**: Controls where climate stress lands (which sectors/regions) and with what sign

(B2) Driver strength scalars λ_k (if using structured W) - **Knob**: λ_k - **Effect**: Overall magnitude of impact of driver k on credit factors - **Calibration style**: "Reasonableness targets" (most exposed cells move ~1–2σ under severe scenario)

(B3) Intercept α - **Knob**: $\alpha_{s,r}$ (usually set to 0) - **Effect**: Baseline offset; cleaner to set $\alpha=0$ and let baseline be zero-stress state

C) Residual credit correlation knobs

(C1) Sector correlation Corr_S - **Knob**: (S×S) correlation matrix - **Effect**: Controls how different sectors co-move - **Calibration**: Estimated from MSCI sector indices (e.g., 5-year rolling correlations)

(C2) Region correlation Corr_R - **Knob**: (R×R) correlation matrix - **Effect**: Controls how different regions co-move - **Calibration**: Estimated from MSCI regional indices

(C3) Volatility scales D_S, D_R - **Knob**: Diagonal matrices - **Effect**: Overall strength of sector/region components in residual systematic risk - **Start**: D_S = D_R = I (unit volatility)

(C4) Sector-vs-region mix ω - **Knob**: $\omega \in [0,1]$ - **Effect**: Controls whether residual co-movement is sector-driven or region-driven - **Interpretation**: - $\omega = 1$: Pure sector co-movement - $\omega = 0$: Pure regional co-movement - $\omega = 0.5$: Balanced - **Typical range**: $\omega \in [0.3, 0.7]$ - **Calibration**: Start with 0.5, adjust based on portfolio sector vs geographic concentration

(C5) Structured-vs-cell share η - **Knob**: $\eta \in [0,1]$ - **Effect**: Controls how much residual variance is structured vs cell-specific - **Interpretation**: - $\eta = 1$: All variance is structured (maximum clustering) - $\eta = 0$: All variance is cell-specific (maximum diversification) - $\eta = 0.7$: Moderately systemic (70% structured, 30% cell-specific) - **Typical range**: $\eta \in [0.5, 0.9]$ - **Calibration**: Start with 0.7, adjust to match portfolio tail thickness and diversification behavior

D) Obligor-level "systematic strength" knob

(D1) Global ϱ - **Knob**: $\varrho \in (0,1)$ - **Effect**: Determines how much of each obligor's latent credit index is systematic vs idiosyncratic - Higher $\varrho \Rightarrow$ more clustering, fatter tails - Lower $\varrho \Rightarrow$ more independence, thinner tails - **Typical range**: $\varrho \in [0.15, 0.25]$ - **Extension**: Later, can be sector-specific ϱ_s if needed

E) Credit inputs (portfolio primitives)

These are standard: - **PD_i**: Baseline unconditional PD (horizon-specific) - **LGD_i**: Loss given default - **EAD_i**: Exposure at default - **β_i**: Exposure weights across sector–region cells (sum to 1)

F) Simulation and reporting knobs

- **# simulations**: N_sim (accuracy vs runtime)
- **Confidence levels**: VaR/ES percentile (e.g., 99%, 99.9%)
- **Decomposition granularity**: Sector-only vs sector×region contributions

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9. Key design choices we made (and why)

Choice 1: Cell-level factors $F_{s,r}$

We use a **sector-region cell matrix** $F \in \mathbb{R}^{(S \times R)}$: - **Natural interpretation**: $F_{coal,europe}$ directly represents "coal in Europe" - **Flexible loadings**: Obligors can have multi-cell exposures (diversified revenue) - **Direct climate mapping**: Each cell responds to climate drivers

Choice 2: Decomposition into climate mean + residual

Climate mean $m(X)$: Deterministic, scenario-driven - Selecting "NGFS Net Zero 2050" fixes $m(X)$ completely - Linear mapping: $m(X) = \alpha + W \phi(X)$ - Matches stress testing convention: scenarios are narratives, not forecasts

Stochastic residual $u \sim N(0, \Sigma_u)$: Random, captures non-climate systematic risk - Constructed from market-observable sector/region correlations - $\Sigma_u = \eta \cdot [\omega \cdot (A \Sigma_S A^T) + (1-\omega) \cdot (B \Sigma_R B^T)] + (1-\eta) \cdot I$ - Grounded in equity market data, not climate-credit speculation - Two intuitive knobs: ω (sector vs region) and η (structured vs cell-specific)

This separation makes uncertainty quantification clean: scenario risk (deterministic X) vs. conditional risk (random u).

Choice 3: Market-calibrated Σ_u with intuitive mixing parameters

Key innovation: We **cannot estimate climate-credit correlations** from historical data.

Instead: - Estimate $Corr_S$ and $Corr_R$ from **observable equity/credit market data** (MSCI indices) - Mix sector and region components with ω (sector-vs-region balance) - Mix structured vs cell-specific with η (clustering strength) - Only 4 key knobs: $Corr_S$, $Corr_R$, ω, η (instead of 5,460 covariance parameters)

Benefits: - **Empirically grounded**: Uses 20+ years of equity market data - **Intuitive knobs**: ω and η have clear economic interpretations - **Separable**: Can tune sector/region balance independently from clustering strength - **Stable**: Well-conditioned by construction, positive definite - **Governance-friendly**: Easy to explain and validate

Choice 4: Single ϱ parameter (asset correlation)

Rather than let each obligor have different systematic loading strength, we: - Standardize systematic scores: $\tilde{S}_i = S_i / \sqrt{(\beta_i^T \Sigma_u \beta_i)}$ - Apply single global ϱ: $Z_i = \sqrt{\varrho}\, \tilde{S}_i + \sqrt{(1-\varrho)}\, \varepsilon_i$

Benefits: - **Simplicity**: One global knob for tail thickness - **Familiarity**: ϱ is standard in Basel/internal models - **Sensitivity**: Easy to report results for $\varrho \in [0.15, 0.25]$ range

Choice 5: Climate affects credit only through systematic factors

No direct climate term in Z_i. All climate impact flows through f:

```
Z_i = √ρ · (β_i^T f / √(β_i^T Σ_u β_i)) + √(1−ρ) · ε_i
where f = m(X) + u
```

Implications: - Climate does not create within-cell differentiation (unless β differs) - Cross-sectional variation comes from sector-region assignments - All climate-driven correlation flows through systematic factors

This is appropriate for portfolio-level stress testing where obligor-specific climate data is sparse.

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10. Calibration workflow (practical "start building" plan)

Step 1: Define taxonomy

- Choose S sectors and R macro-regions
- Define mapping from obligors to β_i weights (summing to 1)

Step 2: Choose climate driver set X and scaling σ_k

- Start with a small global driver set (3–6)
- Define σ_k for each driver (cross-scenario standard deviation)

Step 3: Specify climate-to-credit mapping

- Either define W directly or use structured $S_{s,k}$, $R_{r,k}$ and calibrate λ_k
- Set $\alpha = 0$ (baseline is zero-stress state)

Step 4: Build residual covariance Σ_u

- **Download MSCI sector indices** (15 sectors, 5-year rolling returns)
- **Compute Corr_S** (15×15 sector correlation matrix)
- **Download MSCI regional indices** (7 regions, 5-year rolling returns)
- **Compute Corr_R** (7×7 region correlation matrix)
- Apply shrinkage if needed for stability (e.g., Ledoit-Wolf)
- Set $D_S = D_R = I$ (unit volatility)
- **Choose ω** (sector-vs-region mix, start with 0.5)
- **Choose η** (structured-vs-cell share, start with 0.7)
- Construct membership matrices A (M×S) and B (M×R)
- Construct $\Sigma_u = \eta \cdot [\omega \cdot (A \Sigma_S A^T) + (1-\omega) \cdot (B \Sigma_R B^T)] + (1-\eta) \cdot I$

Step 5: Choose ϱ

- Pick a single global ϱ (start with 0.20)
- Run sensitivity bands: $\varrho \in [0.15, 0.25]$

Step 6: Run Monte Carlo and report

- Baseline and scenario loss distributions
- Contributions by sector and region
- Sensitivities to ϱ, ω, η, and λ_k

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11. Simulation algorithm (single horizon)

Given scenario $X = \mu^{(scenario)}$:

1. Compute features $\phi(X)$
2. Compute deterministic mean $m(X) = \alpha + W\,\phi(X)$
3. Draw residual systematic factor $u \sim N(0, \Sigma_u)$
4. Form $f = m(X) + u$
5. For each obligor i:
    - Compute systematic score $S_i = \beta_i^T f$
    - Standardize $\tilde{S}_i = S_i / \sqrt{(\beta_i^T \Sigma_u \beta_i)}$
    - Draw $\varepsilon_i \sim N(0,1)$
    - Compute $Z_i = \sqrt{\varrho}\,\tilde{S}_i + \sqrt{(1-\varrho)}\,\varepsilon_i$
    - Set $Default_i = 1\{Z_i < \Phi^{(-1)}(PD_i)\}$
    - Compute $L_i = Default_i \cdot EAD_i \cdot LGD_i$
6. Aggregate $L = \Sigma_i L_i$

Repeat (3)–(6) to obtain the loss distribution under the scenario.

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12. What this model can and cannot claim

What it can do well

- Produce coherent loss distributions under climate scenarios.
- Attribute incremental risk to sector and region segments in a controlled way.
- Provide transparent, explainable linkage from scenario variables to credit stress.
- Support stress testing, risk appetite discussion, and portfolio steering.
- Use market-observable data (sector/region correlations) rather than speculative climate-credit correlations.

What it should not claim (without further development)

- Precise prediction of obligor defaults driven by climate.
- Fine-grained within-sector differentiation unless β_i includes richer exposure splits.
- Causal inference from historical climate to credit outcomes.

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13. One-paragraph “model story” (for stakeholders)

We take climate scenarios as deterministic narratives expressed through a small set of global driver variables X. These drivers shift the mean systematic credit conditions for each sector–region cell via a transparent mapping m(X). Around that scenario-implied mean, we add a stochastic residual systematic credit environment u whose correlations across sectors and regions are calibrated using observable market co-movement (sector and macro-region indices). Obligors load on the sector–region surface via exposure weights β_i, and a single parameter ϱ controls how strongly obligors co-move with systematic conditions versus idiosyncratic noise. This produces coherent loss distributions under scenarios without claiming to estimate causal climate–default relationships from limited historical data.

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14. Notation Reference

Symbol	Description
X	Deterministic climate/scenario variable vector
F	Sector–region factor matrix (systematic credit conditions)
$m(X)$	Deterministic scenario-driven mean of factor surface
u	Stochastic residual $\sim N(0, \Sigma_u)$
Σ_u	Residual covariance matrix ($M \times M$)
$Corr_S$	Sector correlation matrix ($S \times S$), estimated from MSCI sector indices
$Corr_R$	Region correlation matrix ($R \times R$), estimated from MSCI regional indices
A	Membership matrix ($M \times S$), maps cells to sectors
B	Membership matrix ($M \times R$), maps cells to regions
ω	Sector-vs-region mix parameter $\in [0,1]$ ($\omega=1$: all sector, $\omega=0$: all region)
η	Structured-vs-cell share parameter $\in [0,1]$ ($\eta=1$: all structured, $\eta=0$: all cell-specific)
ϱ	Asset correlation (fraction of variance that is systematic)
Z_i	Latent creditworthiness index for obligor i
S_i	Systematic score for obligor i: $\beta_i^T f$
$\tilde{S}_i$	Standardized systematic score: $S_i / \sqrt{(\beta_i^T \Sigma_u \beta_i)}$
β_i	Factor loadings/exposures for obligor i (sum to 1)
ε_i	Idiosyncratic noise for obligor i $\sim N(0,1)$
τ_i	Default threshold for obligor i: $\Phi^{(-1)}(PD_i)$
Φ	Standard normal cumulative distribution function
PD_i	Probability of default for obligor i

Symbol	Description
LGD_i	Loss given default for obligor i
EAD_i	Exposure at default for obligor i
W	Sensitivity matrix ($M \times K$)
$\phi(X)$	Standardized feature transformation of climate variables
$S_{s,k}$	Sector exposure score for sector s to driver k (optional hybrid structure)
$R_{r,k}$	Region exposure score for region r to driver k (optional hybrid structure)
λ_k	Scale parameter for driver k (optional hybrid structure)
σ_k	Standardization scale for driver k